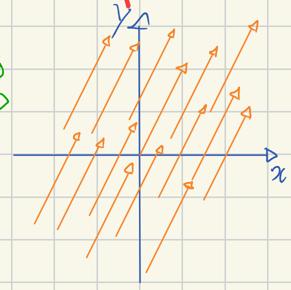
Lecture 19: Vector fields and line integrals in the plane

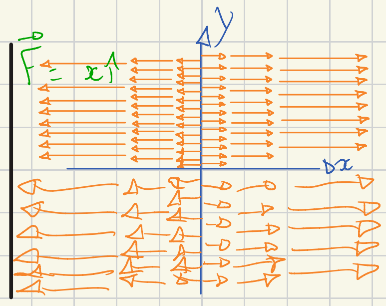
Vector fields: $\vec{F} = M\hat{\imath} + N\hat{\jmath}$, M and N are functions of x, y.
(Example: Wind Maps)

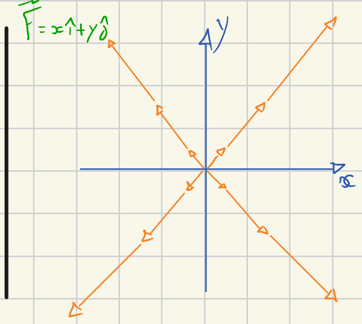
At each point (x,y), $\vec{F}$ a vector that depends on (x,y).

Examples: * velocity in a fluid $\vec{v}$
* force field $\vec{F}$

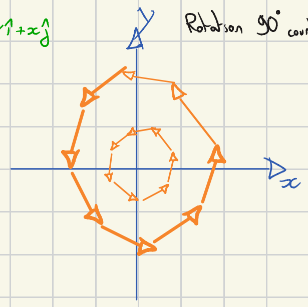
Examples: $\vec{F} = 2\hat{\imath} + \hat{\jmath}$
$\langle 2;1 \rangle$



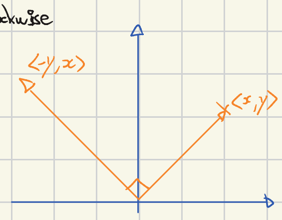




$\vec{F} = -y\hat{\imath} + x\hat{\jmath}$

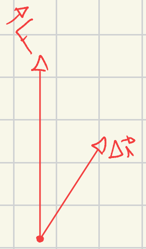


Rotation 90° counterclockwise



Velocity field for uniform rotation at unit angular velocity

Work and line integrals: $W = (\text{force}) \cdot (\text{distance}) = \vec{F} \cdot \Delta\vec{r}$

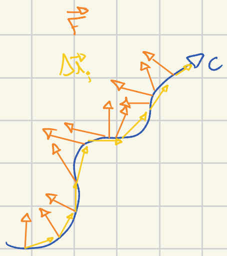


Along some trajectory C, work adds up to:

$$W = \int_C \vec{F} \cdot d\vec{r} \quad \left(\Leftrightarrow \lim_{\Delta r \to 0} \sum_i \vec{F} \cdot \Delta\vec{r}_i \right)$$

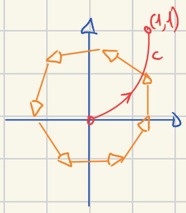
$$\left(\Leftrightarrow \sum_i \vec{F} \cdot \left(\underbrace{\frac{\Delta\vec{r}}{\Delta t}}_{\text{velocity vector } \frac{d\vec{r}}{dt}} \Delta t \right) \right)$$

$$(\Rightarrow W = \int_{t_1}^{t_2} \vec{F} \cdot \frac{d\vec{r}}{dt} \, dt$$



Example: $\vec{F} = -y\hat{\imath} + x\hat{\jmath}$ $\quad C: x = t, \ y = t^2 \quad 0 \leq t \leq 1$

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 \vec{F} \cdot \frac{d\vec{r}}{dt} \, dt = \int_0^1 \overbrace{\langle -t^2, t \rangle}^{\vec{F}} \cdot \overbrace{\langle 1, 2t \rangle}^{\frac{d\vec{r}}{dt}} dt = \int_0^1 (-t^2 - 2t^2)\,dt = \int_0^1 t^2 \, dt = \frac{1}{3} = W$$



$\vec{F} = \langle -y, x \rangle \Leftrightarrow \langle -t^2, t \rangle$
$\begin{cases} x = t \\ y = t^2 \end{cases}$ $\quad dx/dt = 1$, $dy/dt = 2$

Another Way: $\vec{F} = \langle M, N \rangle$ $\quad \vec{F} \cdot d\vec{r} = M\,dx + N\,dy$
$d\vec{r} = \langle dx, dy \rangle$

$\int_C \vec{F} \cdot d\vec{r} = \int_C M\,dx + N\,dy$ $\qquad$ Method to evaluate: express x, y in terms of a single variable & substitute

$\int_C \vec{F} \cdot d\vec{r} = \int_C -y\,dx + x\,dy$ = in terms of t with $x = t, y = t^2$, $dx = dt$, $dy = 2t\,dt$ $\Leftrightarrow \int_C -t^2\,dt + t \cdot 2t\,dt = \int_0^1 t^2\,dt = \frac{1}{3}$

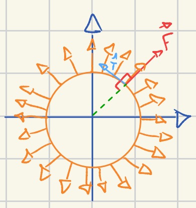
Note: $\int_C \vec{F} \cdot d\vec{r}$ depends on the trajectory c but not on parametrisation. e.g. could do = $\begin{cases} x = \sin\theta \\ y = \sin^2\theta \end{cases}$ $\quad 0 \leq \theta \leq \frac{\pi}{2}$

Geometric Approach:

$d\vec{r} = \langle dx, dy \rangle = \hat{T}\,ds$

Note: $\frac{d\vec{r}}{dt} = \langle \frac{dx}{dt}, \frac{dy}{dt} \rangle = \hat{T} \underbrace{\frac{ds}{dt}}_{\text{speed}}$ $\qquad$ So $\int_C \vec{F} \cdot d\vec{r} = \int_C M\,dx + N\,dy = \int_C \vec{F} \cdot \hat{T}\,ds$

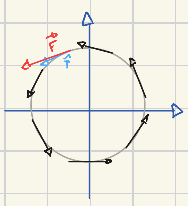
Example: C: circle of radius a at origin, counterclockwise, $\vec{F} = x\hat{\imath} + y\hat{\jmath}$



$\vec{F} \perp \hat{T} = 0$, $\int_C \vec{F} \cdot \hat{T}\,ds = 0$
$\vec{F} \cdot \hat{T} = 0$

Example 2: same c, $\vec{F} = -y\hat{\imath} + x\hat{\jmath}$

Now $\vec{F} \parallel \hat{T}$, $\vec{F} \cdot \hat{T} = |\vec{F}| = a$



$$\int_C \vec{F} \cdot \hat{T}\,ds = \int_C a\,ds = a \int_C ds = a \underbrace{\text{length}(c)}_{2\pi a} = 2\pi a^2$$

Or computing: $\int_C -y\,dx + x\,dy$ with $x = a\cos\theta$, $y = a\sin\theta$ $\quad 0 \leq \theta \leq 2\pi$

$$\| $$

$$\int_C -(a\sin\theta)(-a\sin\theta\,d\theta) + (a\cos\theta)(a\cos\theta\,d\theta)$$

$$= \int_0^{2\pi} a^2 \underbrace{(\sin^2\theta + \cos^2\theta)}_{=1} d\theta = 2\pi a^2$$